

E025

Cloud and Network Security Models

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A marketing agency runs its client portal on IaaS virtual machines from a large provider that holds many compliance certifications. A breach traced back to a guest operating system patch that had been available for four months. The agency has 85 staff. Under shared responsibility, who should have applied the patch?

- A. The provider, because it owns the hardware.
- B. Neither; patching is optional in the cloud.
- C. The agency.
- D. The provider, because its certifications cover patching.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which security task belongs to the cloud provider in every service model (IaaS, PaaS, and SaaS)?

- A. Classifying customer data.
- B. Physical security of the data centers.
- C. Managing customer user accounts.
- D. Patching the guest operating system.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A regional hospital lets any device on its internal network reach the medical-records application without further checks. The network was rebuilt last year with new switches, and staff badges are required to enter the building. Which change best applies zero trust?

- A. Rely on building badges, since only staff can reach the network.
- B. Put everything behind a VPN, which makes it zero trust.
- C. Buy faster switches.
- D. Require each request to verify user identity, device health, and least-privilege authorization.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Under a zero-trust model, a laptop that is already connected to the corporate VPN requests access to a finance application. What should happen?

- A.** Allow it if the VPN is encrypted.
- B.** Allow it, since VPN users are trusted.
- C.** Deny it, since zero trust means no access.
- D.** Verify the user and device and check authorization for that app before granting access.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A logistics company protects its cloud workloads with a well-tuned perimeter firewall. Workloads share one flat subnet, admin accounts have no MFA, and logs are not collected. The firewall rules were reviewed by an outside firm in May. What is the key weakness?

- A.** The firewall review is out of date.
- B.** Nothing; a well-tuned firewall stops all attacks.
- C.** No depth: a single bypass of the firewall exposes everything, with no identity, segmentation, or detection layers.
- D.** The firewall should be replaced with a stronger one.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A cloud team replaces a stateful security group with a stateless network ACL on a web subnet. They allow inbound HTTPS on port 443 but add no outbound rules. The subnet serves about 2,000 users an hour. Users now report that pages never load. Why?

- A.** The stateless ACL blocks the return traffic because no outbound rule allows it.
- B.** The subnet has too many users.
- C.** Stateless filters cannot allow HTTPS at all.
- D.** Port 443 is the wrong port for HTTPS.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A stateful cloud firewall allows inbound HTTPS to a web server. Does it need an explicit outbound rule for the server's replies to those requests?

- A.** Yes; every direction always needs its own rule.
- B.** No, because stateful firewalls allow all outbound traffic.
- C.** Yes, but only for encrypted traffic.
- D.** No; return traffic for allowed connections is permitted automatically.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A law firm encrypts client files at rest in cloud storage using provider-managed keys. A paralegal with valid read access downloads files she should not see. The storage uses the provider's newest, fastest tier. What does at-rest encryption do in this case?

- A.** It blocks her, because the files are encrypted.
- B.** Nothing to stop her; the fix is least-privilege access and monitoring.
- C.** It blocks her only if the fastest tier is used.
- D.** Switching to customer-managed keys would stop her.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A developer at a 300-person software firm stores a cloud access key with full administrator rights in a code repository so a reporting script can read one storage bucket. The script runs every night at 2 a.m. What is the best fix?

- A.** Give every developer the same admin key so access is consistent.
- B.** Run the script at a different time.
- C.** Remove and rotate the key; give the script a role with read-only access to that one bucket and short-lived credentials.
- D.** Keep the key but make the repository private.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A retailer passed a cloud security audit in January. In June, an engineer made a storage bucket public to share a file with a partner and forgot to revert it. The retailer's cloud bill is \$90,000 a month. What control would most likely have caught this?

- A.** The January audit, which proved the environment secure.
- B.** Continuous cloud security posture management with alerts on public exposure.
- C.** A stronger perimeter firewall.
- D.** A lower cloud bill.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A bank connects its data center to its cloud environment through a dedicated private circuit leased from a carrier for \$12,000 a month. A risk officer asks whether data on the circuit is encrypted. The carrier has served the bank for ten years. What is the correct answer?

- A.** Not by default; the bank should add encryption (for example, IPsec or application TLS) if required.
- B.** It does not matter, since a VPN would be slower.
- C.** Yes, because the carrier is trusted.
- D.** Yes, because private circuits are always encrypted.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A small branch office needs an encrypted connection to the company's cloud network quickly and at low cost, using its existing internet service. Which option fits?

- A.** A site-to-site VPN.
- B.** Unencrypted access over the public internet.
- C.** A dedicated private circuit.
- D.** A second internet connection.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A marketing agency runs its client portal on IaaS virtual machines ~~from a large provider that holds many compliance certifications~~. A breach traced back to a guest operating system patch that had been available for four months. ~~The agency has 85 staff~~. Under shared responsibility, who should have applied the patch?

Target word: owner of guest OS patching in IaaS **Decides:** In IaaS the customer manages the guest OS.

A	Adjacent category	Hardware and hypervisor are the provider's; the guest OS is not.
B	Absolute language	Patching remains a core customer duty.
C	Correct	The customer owns the OS and everything above it in IaaS.
D	Camouflage	Provider certifications cover its layers, not the customer's VMs.

2. Answer B Concept 1 • transfer

Which security task belongs to the cloud provider in every service model (IaaS, PaaS, and SaaS)?

Target word: reverse: find the always-provider duty **Decides:** Physical and hypervisor layers are always the provider's.

A	Confusion set	Always the customer's.
B	Correct	Always the provider's responsibility.
C	Inverted operation	Always the customer's.
D	Adjacent category	Customer's in IaaS; shifts only in PaaS and SaaS.

3. Answer D Concept 2

A regional hospital lets any device on its internal network reach the medical-records application without further checks. ~~The network was rebuilt last year with new switches~~, and staff badges are required to enter the building. Which change best applies zero trust?

Target word: zero-trust change **Decides:** Access is granted by network location alone.

A	Inverted operation	Doubles down on location-based trust.
B	Adjacent category	A VPN is still a trusted-network model.
C	Camouflage	Hardware speed does not change the trust model.
D	Correct	Removes implicit trust from location.

4. Answer D Concept 2 • transfer

Under a zero-trust model, a laptop that is already connected to the corporate VPN requests access to a finance application. What should happen?

Target word: reverse: "inside" request **Decides:** Being on the VPN earns no automatic trust.

A	Confusion set	Encryption protects transit, not authorization.
B	Inverted operation	That is implicit network trust.
C	Absolute language	Zero trust grants verified, least-privilege access.
D	Correct	Every request is evaluated on its own.

5. Answer C Concept 3

A logistics company protects its cloud workloads with a well-tuned perimeter firewall. Workloads share one flat subnet, admin accounts have no MFA, and logs are not collected. ~~The firewall rules were reviewed by an outside firm in May.~~ What is the key weakness?

Target word: weakness in the model **Decides:** One layer, nothing behind it.

A	Camouflage	The review does not decide; missing layers do.
B	Absolute language	No single control stops everything.
C	Correct	Defense in depth needs independent layers.
D	Inverted operation	Strengthens the single layer instead of adding others.

6. Answer A Concept 4

A cloud team replaces a stateful security group with a stateless network ACL on a web subnet. They allow inbound HTTPS on port 443 but add no outbound rules. ~~The subnet serves about 2,000 users an hour.~~ Users now report that pages never load. Why?

Target word: cause of failed responses **Decides:** Stateless filters need explicit rules for return traffic.

A	Correct	Stateless filters do not track connections.
B	Camouflage	Load does not decide this failure.
C	Absolute language	They can, with rules in both directions.
D	Confusion set	443 is correct.

7. Answer D Concept 4 · transfer

A stateful cloud firewall allows inbound HTTPS to a web server. Does it need an explicit outbound rule for the server's replies to those requests?

Target word: reverse: stateful behavior **Decides:** Stateful filters track sessions.

A	Inverted operation	That is stateless behavior.
B	Adjacent category	Replies are allowed; unrelated new outbound flows may still be filtered.
C	Confusion set	Encryption does not affect state tracking.
D	Correct	Stateful filters remember the session.

8. Answer B Concept 5

A law firm encrypts client files at rest in cloud storage using provider-managed keys. A paralegal with valid read access downloads files she should not see. ~~The storage uses the provider's newest, fastest tier.~~ What does at-rest encryption do in this case?

Target word: limits of at-rest encryption **Decides:** Authorized users see decrypted data.

A	Inverted operation	Storage decrypts transparently for authorized users.
B	Correct	Encryption at rest protects media, not against valid access.
C	Camouflage	Storage tier does not matter.
D	Adjacent category	Key custody controls who can decrypt at the provider level, not an already-authorized user.

9. Answer C Concept 6

A developer at a 300-person software firm stores a cloud access key with full administrator rights in a code repository so a reporting script can read one storage bucket. ~~The script runs every night at 2 a.m.~~ What is the best fix?

Target word: identity fix **Decides:** Long-lived, over-privileged key in code.

A	Inverted operation	Spreads the risk.
B	Camouflage	Timing does not change exposure.
C	Correct	Least privilege and no embedded long-lived keys.
D	Dropped element	Leaves the over-privileged, long-lived key in place.

10. Answer B Concept 7

A retailer passed a cloud security audit in January. In June, an engineer made a storage bucket public to share a file with a partner and forgot to revert it. ~~The retailer's cloud bill is \$90,000 a month.~~ What control would most likely have caught this?

Target word: control for configuration drift **Decides:** A change after the audit created exposure.

A	Absolute language	A point-in-time audit does not cover later changes.
B	Correct	Detects drift as it happens.
C	Adjacent category	Public storage is reached through the provider's service, not the customer perimeter.
D	Camouflage	Spend does not decide exposure.

11. Answer A Concept 8

A bank connects its data center to its cloud environment through a dedicated private circuit leased from a carrier for ~~\$12,000 a month~~. A risk officer asks whether data on the circuit is encrypted. ~~The carrier has served the bank for ten years.~~ What is the correct answer?

Target word: encryption on a private circuit **Decides:** Private does not mean encrypted.

A	Correct	Dedicated circuits are private but not automatically encrypted.
B	Confusion set	Performance does not answer the encryption question.
C	Camouflage	Trust in a carrier is not a cryptographic control.
D	Absolute language	Privacy of the path is not encryption.

12. Answer A Concept 8 • transfer

A small branch office needs an encrypted connection to the company's cloud network quickly and at low cost, using its existing internet service. Which option fits?

Target word: reverse: choose the connection **Decides:** Encrypted, fast to set up, uses the internet.

A	Correct	Encrypts traffic over the existing internet link.
B	Dropped element	Drops the encryption requirement.
C	Inverted operation	Costly, slower to provision, and not encrypted by default.
D	Adjacent category	Adds bandwidth, not encryption.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	2	
Wrong stem number	A figure copied from the prompt that is not the answer.	0	
Adjacent category	The right idea in the wrong bucket.	7	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	5	
Camouflage	Answers a stem fact that was never the question.	8	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Shared responsibility for security	The provider secures the cloud (facilities, hardware, hypervisor); the customer secures what it puts in the cloud — in IaaS that includes the guest OS, network rules, identities, and data.	Assuming the provider patches customer VMs or configures customer firewalls.	A security gap in a customer-managed layer.
2	Zero trust	Grant no implicit trust based on network location; verify every request by identity, device health, and context, with least privilege.	Trusting anything inside the corporate network or VPN.	“Inside the network, so it is trusted.”
3	Defense in depth	Layer independent controls (identity, network, host, application, data, monitoring) so one failure does not expose the asset.	Relying on a single perimeter firewall.	One strong control and nothing behind it.
4	Stateful vs. stateless filtering	Stateful filters (e.g., cloud security groups) track connections and allow return traffic automatically; stateless filters (e.g., network ACLs) evaluate each packet and need explicit rules both ways.	Forgetting return-traffic rules on a stateless filter, or adding them needlessly on a stateful one.	Outbound works but replies are dropped, or rule sets for each direction.
5	Encryption and key control	Encrypt data in transit and at rest; who controls the keys determines who can read the data — customer-managed keys keep that control with the customer.	Believing encryption at rest protects data from a user with valid access.	A question about who can decrypt or about key custody.
6	Identity as the control plane	Use least privilege, MFA, and short-lived role-based credentials; avoid long-lived keys embedded in code.	Granting broad admin rights for convenience.	An account or key with more access than its task needs.
7	Misconfiguration and posture management	Most cloud breaches trace to customer misconfiguration; continuous posture management and centralized logging catch drift and exposure.	Treating a one-time audit as lasting assurance.	A storage bucket or rule made public after a change.
8	Hybrid connectivity: VPN vs. private link	Site-to-site VPN encrypts traffic over the public internet; a dedicated private circuit gives predictable bandwidth and latency but is private, not automatically encrypted.	Assuming a dedicated circuit is encrypted by default.	A choice between internet VPN and a leased private connection.

Course scope (public description): Cloud and Network Security Models guides learners through the intricacies of cloud and hybrid network security, covering fundamental principles, emerging threats, cutting-edge tools, advanced defense techniques and best practices for fortifying network security and mitigating risks.